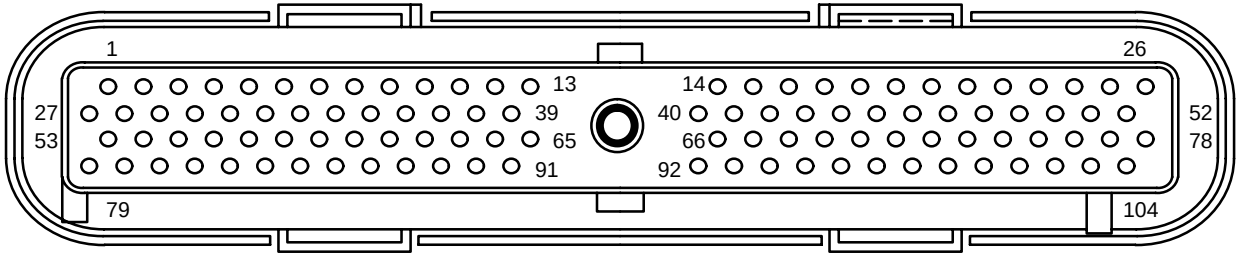
## C175

**12A581**

A147
Powertrain Control Module (PCM) (12A650)




F104001

| Pin | Circuit | Circuit function |
|---|---|---|
| 1 | 1025 (OG/YE) | Coil On Plug (COP) 6 (12029), control |
| 2 | – | not used |
| 3 | 570 (BK/WH) | Ground |
| 4 | – | not used |
| 5 | – | not used |
| 6 | 237 (OG/YE) | Shift solenoid A |
| 7 | – | not used |
| 8 | – | not used |
| 9 | – | not used |
| 10 | – | not used |
| 11 | 315 (VT/OG) | Shift solenoid B |
| 12 | 911 (WH/LG) | Transmission Control Indicator Lamp (TCIL) |
| 13 | 107 (VT) | Module programming signal |
| 14 | – | not used |
| 15 | 915 (PK/LB) | SCP Bus – |
| 16 | 914 (TN/OG) | SCP Bus + |
| 17 | – | not used |
| 18 | – | not used |
| 19 | 637 (DB/WH) | Air suspension module (3C142), to, Powertrain Control Module (PCM) (12A650) |
| 20 | 924 (BN/OG) | Powertrain Control Module (PCM) (12A650), to transmission coast clutch solenoid |

| Pin | Circuit | Circuit function |
|---|---|---|
| 21 | 349 (DB) | Crankshaft position sensor (6C315) + |
| 22 | 350 (GY) | Crankshaft position sensor (6C315) – |
| 23 | – | not used |
| 24 | – | not used |
| 25 | 567 (LB/YE) | Power ground |
| 26 | 1024 (LG/WH) | Coil On Plug (COP) 1 (12029), control |
| 27 | 1021 (LG/YE) | Coil On Plug (COP) 5 (12029), control |
| 28 | – | not used |
| 29 | 224 (TN/WH) | PCM, to transmission overdrive cancel switch |
| 30 | – | not used |
| 31 | – | not used |
| 32 | 311 (DG/VT) | Knock sensor (12A699) |
| 33 | – | not used |
| 34 | 1144 (YE/BK) | Digital Transmission Range (DTR) sensor, TR1 |
| 35 | 392 (RD/LG) | Heated Oxygen Sensor (HO2S) #12 (9G444) |
| 36 | 968 (TN/LB) | Mass Air Flow (MAF) sensor (12B579), signal, return |
| 37 | 923 (OG/BK) | Transmission Fluid Temperature (TFT) sensor, input |
| 38 | 354 (LG/RD) | Engine Coolant Temperature (ECT) sensor (12A648), input |
| 39 | 743 (GY) | Intake Air Temperature (IAT) sensor (12A697), input |
| 40 | 238 (DG/YE) | Fuel pump relay, switched power, output |
| 41 | 347 (BK/YE) | A/C clutch relay, switched output |
| 42 | – | not used |

| Pin | Circuit | Circuit function |
|---|---|---|
| 43 | 71 (OG/LG) | Powertrain Control Module (PCM) (12A650) – Data output link |
| 44 | – | not used |
| 45 | – | not used |
| 46 | 367 (BN) | Intake Manifold Runner Control (IMRC) module |
| 47 | 360 (BN/PK) | EGR Vacuum Regulator (EVR), control |
| 48 | – | not used |
| 49 | 1145 (LB/BK) | Digital Transmission Range (DTR) sensor, TR2 |
| 50 | 1143 (WH/BK) | Digital Transmission Range (DTR) sensor, TR4 |
| 51 | 570 (BK/WH) | Ground |
| 52 | 1028 (WH/PK) | Coil On Plug (COP) 3 (12029), control |
| 53 | 1030 (DG/VT) | Coil On Plug (COP) 4 (12029), control |
| 54 | 480 (VT/YE) | Torque Converter Clutch (TCC) solenoid |
| 55 | 729 (RD/WH) | Voltage supplied at all times (overload protected) |
| 56 | 191 (LG/BK) | Vapor management valve |
| 57 | 310 (YE/RD) | Knock sensor (12A699) |
| 58 | – | not used |
| 59 | 970 (DG/WH) | Turbine shaft speed sensor (7M101) |
| 60 | 74 (GY/LB) | Heated Oxygen Sensor (HO2S) #11 (9F472) |
| 61 | 393 (VT/LG) | Heated Oxygen Sensor (HO2S) #22 (9G444), input |
| 62 | 791 (RD/PK) | Fuel tank pressure transducer sensor (9C968) |
| 63 | – | not used |
| 64 | 199 (LB/YE) | Digital Transmission Range (DTR) sensor, to PCM |
| 65 | 352 (BN/LG) | Differential Pressure Feedback EGR (DPFE) sensor, input |
| 66 | 1102 (YE/LG) | Cylinder–head temperature sensor (6G004), to, Powertrain Control Module (PCM) (12A650) |
| 67 | 91 (VT/WH) | Canister vent |
| 68 | 679 (GY/BK) | Vehicle Speed Sensor (VSS) + |
| 69 | 331 (PK/YE) | A/C clutch relay, control |
| 70 | – | not used |
| 71 | 361 (RD) | Voltage supplied in Start and Run (overload protected) |
| 72 | 561 (TN/RD) | Injector 7, control |
| 73 | 559 (TN/BK) | Injector 5, control |

| Pin | Circuit | Circuit function |
|---|---|---|
| 74 | 557 (BN/YE) | Injector 3, control |
| 75 | 555 (TN) | Injector 1, control |
| 76 | 570 (BK/WH) | Ground |
| 77 | – | not used |
| 78 | 1027 (PK/LB) | Coil On Plug (COP) 7 (12029), control |
| 79 | 1029 (WH/RD) | Coil On Plug (COP) 8 (12029), control |
| 80 | 926 (LB/OG) | Fuel pump relay, control |
| 81 | 925 (WH/YE) | to Electronic Pressure Control (EPC) solenoid |
| 82 | – | not used |
| 83 | 264 (WH/LB) | Idle air control valve (IAC) (9F715), control |
| 84 | 136 (DB/YE) | Powertrain Control Module (PCM) (12A650) – Output Shaft Speed (OSS) sensor (7M101) |
| 85 | 795 (DG) | Camshaft position sensor (6B288), to, Powertrain Control Module (PCM) (12A650) |
| 86 | – | not used |
| 87 | 94 (RD/BK) | Heated Oxygen Sensor (HO2S) #21 (9F472) |
| 88 | 967 (LB/RD) | Mass Air Flow (MAF) sensor (12B579) |
| 89 | 355 (GY/WH) | Throttle Position Sensor (TPS) (9B989), input |
| 90 | 351 (BN/WH) | Reference, voltage |
| 91 | 359 (GY/RD) | signal, return |
| 92 | 810 (RD/LG) | Brake pedal position switch (13480), input |
| 93 | 387 (RD/WH) | Heated Oxygen Sensor (HO2S) #11 (9F472), heater |
| 94 | 388 (YE/LB) | Heated Oxygen Sensor (HO2S) #21 (9F472), heater |
| 95 | 389 (WH/BK) | Heated Oxygen Sensor (HO2S) #12 (9G444), heater |
| 96 | 390 (TN/YE) | Heated Oxygen Sensor (HO2S) #22 (9G444), heater |
| 97 | 361 (RD) | Voltage supplied in Start and Run (overload protected) |
| 98 | 562 (LB) | Injector 8, control |
| 99 | 560 (LG/OG) | Injector 6, control |
| 100 | 558 (BN/LB) | Injector 4, control |
| 101 | 556 (WH) | Injector 2, control |
| 102 | – | not used |
| 103 | 570 (BK/WH) | Ground |
| 104 | 1026 (PK/WH) | Coil On Plug (COP) 2 (12029), control |